
Online Examination Report

Date:	17.07.2014
Attempts:	1
Examinee:	Ekwee, Simon 2
Date of birth:	8/29/1987
Subject:	020-AGK
Result:	40 %
Time used:	00:30:00 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	2722	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
2	3657	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
3	4494	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
4	2409	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
5	843	21.2.4	Fuselage, landing gear, doors, floor, etc	1,00	1,00
6	25	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
7	546	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
8	2929	21.8	FUEL SYSTEM	1,00	1,00
9	1763	21.8.2.2	Design, operation, system components, etc	0,00	1,00
10	548	21.9.3.1	DC Generation	0,00	1,00
11	96	21.9.3.1	DC Generation	0,00	1,00
12	198	21.10.1.2	Engine: design, operation etc	1,00	1,00
13	671	21.10.10	Performance and engine handling	0,00	1,00
14	1462	21.11.1	Basic principles	0,00	1,00
15	1310	21.11.1.2	Design, types of turbine engines, components	1,00	1,00
16	547	21.11.1.2	Design, types of turbine engines, components	1,00	1,00
17	1164	21.11.2.4	Turbine	0,00	1,00
18	385	21.12.2.2	Fire detection	1,00	1,00
19	1818	21.13	OXYGEN SYSTEMS	1,00	1,00
20	343	21.13	OXYGEN SYSTEMS	0,00	1,00
21	4132	22.	INSTRUMENTATION	1,00	1,00
22	3097	22.	INSTRUMENTATION	0,00	1,00
23	2815	22.	INSTRUMENTATION	1,00	1,00
24	2801	22.	INSTRUMENTATION	0,00	1,00
25	3094	22.	INSTRUMENTATION	0,00	1,00
26	4028	22.	INSTRUMENTATION	0,00	1,00
27	4026	22.	INSTRUMENTATION	1,00	1,00
28	2809	22.	INSTRUMENTATION	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	1123	22.2	MEASUREMENT OF AIR DATA PARAMETERS	0,00	1,00
30	792	22.2.1.1	Definitions	0,00	1,00
31	242	22.2.4	Altimeter	0,00	1,00
32	1910	22.2.4	Altimeter	0,00	1,00
33	65	22.2.6	Airspeed indicator (ASI)	1,00	1,00
34	1294	22.2.7	Mach meter	0,00	1,00
35	1307	22.2.7	Mach meter	0,00	1,00
36	1736	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	0,00	1,00
37	1646	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	0,00	1,00
38	175	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator	0,00	1,00
39	1422	22.6.3	Flight Director : design and operation.	0,00	1,00
40	327	22.12.3	Stall Warning Systems (SWS)	1,00	1,00
Total:40%				16,00	40,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: Ekwee, Simon 2

Location: Uganda Civil Aviation Authority

Your Result: 16,00 from 40,00 Points (40,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	10,00	20,00	50,00
21.2. AIRFRAME	2	1,00	2,00	50,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	2	1,00	2,00	50,00
21.8. FUEL SYSTEM	2	1,00	2,00	50,00
21.9. ELECTRICS	2	0,00	2,00	0,00
21.10. PISTON ENGINES	2	1,00	2,00	50,00
21.11. TURBINE ENGINES	4	2,00	4,00	50,00
21.12. PROTECTION AND DETECTION SYSTEMS	1	1,00	1,00	100,00
21.13. OXYGEN SYSTEMS	2	1,00	2,00	50,00
22. INSTRUMENTATION	20	6,00	20,00	30,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	7	1,00	7,00	14,29
22.3. MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	2	0,00	2,00	0,00
22.4. GYROSCOPIC INSTRUMENTS	1	0,00	1,00	0,00
22.6. AEROPLANE : AUTOMATIC FLIGHT CONTROL SYSTEMS	1	0,00	1,00	0,00
22.12. ALERTING SYSTEMS, PROXIMITY SYSTEMS	1	1,00	1,00	100,00
Total	40	16,00	40,00	40,00

1

When refueling aircraft, which precaution would be adequate for eliminating the potential hazard of static electricity?

- ☒ Connect a ground wire between the aircraft, fuel truck, fuel nozzle, and ground.
- ☐ Connect a ground wire from the fuel truck to ground.
- ☐ Ensure that battery and ignition switches are off.

2

Adding more oil to the fuel than specified by the manufacturer of a 2-cycle engine will result in

- ☐ increased engine performance.
- ☐ increased engine lubrication and optimal performance.
- ☒ increased carbon buildup and engine fouling.

3

After the ignition source is removed, which class of material will not propagate a flame beyond safe limits?

☐ Flame resistant.

☒ Fireproof.

☐ Flash resistant.

4

What is the purpose of an elevator trim tab?

- ☐ Provide horizontal balance as airspeed is increased to allow hands-off flight.
- ☐ Modify the downward tail load for various airspeeds in flight eliminating flight-control pressures.
- ☒ Adjust the speed tail load for different airspeeds in flight allowing neutral control forces.

5

With regard to an aircraft structure, 'fail safe' is one:

- ☐ *that is easily manufactured.*
- ☐ *that is only used for a limited time.*
- ☒ *in which the load is carried by other components if a part of the structure fails.*
- ☐ *used for small aircraft only.*

6

Assuming cabin pressure decreases, the cabin rate of climb indicator should indicate :

- ☐ zero.
- ☒ a rate of climb.
- ☐ a rate of descent dependent upon the cabin differential pressure.
- ☐ a rate of descent of approximately 300 feet per minutes.

7

The term "Bootstrap", when used to identify a cabin air conditioning and pressurisation system, refers to the :

- ☒ cold air unit (air cycle machine) arrangement.
- ☐ source of the charge air.
- ☒ charge air across the inter-cooler heat exchanger.
- ☐ means by which pressurisation is controlled.

8

Why should special precautions be taken when filling propane bottles?

- ☐ During transfer, propane reaches a high temperature and can cause severe burns.
- ☒ Propane is very cold and may cause freeze burns.
- ☐ Propane is under high pressure when transferred from storage tanks to propane bottles.

9

The cross-feed fuel system enables:

- ☒ only the transfer of fuel from the centre tank to the wing tanks.
- ☐ the supply of the outboard jet engines from any outboard
- ☐ the supply of the jet engines mounted on a wing from any fuel tank within that wing.
- ☐ the supply of any jet engine from any fuel tank.

10

The detection of a feeder fault on a direct current circuit results in:

1. automatic disconnection of the generator from the aircraft AC busbar
2. opening of generator field current relay
3. opening of the main relay of the generator breaker
4. opening of balancing circuit connecting two generators
5. lighting of an indicator lamp

The combination of correct statements is:

☐ 2, 4, 5

☐ 2, 3, 4, 5

☒ 1, 2, 3, 4, 5

☐ 1, 3, 5

11

The frequency of the current provided by an alternator depends on...

- ☐ *its phase balance.*
- ☒ *its rotation speed.*
- ☒ *its load.*
- ☐ *the strength of the excitation current.*

12

The part of a piston engine that transforms reciprocating movement into rotary motion is termed as the :

☒ crankshaft

☐ camshaft

☐ reduction gear

☐ piston

13

Which statement is true concerning the effect of the application of carburettor heat?

- ☒ it reduces the density of air entering the carburettor, thus enriching the fuel/air mixture
- ☐ it reduces the volume of air entering the carburettor, thus enriching the fuel/air mixture
- ☒ it reduces the density of air entering the carburettor, thus leaning the fuel/air mixture
- ☐ it reduces the volume of air entering the carburettor, thus leaning the fuel/air mixture

14

In a "bootstrap" cooling system, the charge air is first compressed in the cold air unit to:

- ☐ maintain a constant cabin mass air flow.
- ☒ increase the cabin air supply pressure when the charge pressure is too low.
- ☐ ensure an adequate charge air flow across the inter-cooler heat exchanger.
- ☐ ensure an adequate pressure and temperature drop across the cooling turbine.

15

Turbine blade stages may be classed as either "impulse" or "reaction". In an impulse blade section :

- ☐ the pressure rises across the nozzle guide vanes and remains constant across the rotor blades .
- ☒ the pressure drops across the nozzle guide vanes and remains constant across the rotor blades .
- ☐ the pressure remains constant across the nozzle guide vanes and rises constant across the rotor blades .
- ☐ the pressure remains constant across the nozzle guide vanes and drops across the rotor blades .

16

A turbofan engine with a bypass air mass flow of 200 kg/s and a HP compressor mass flow of 20 kg/s has a bypass ratio of:

☐ 11

☐ 9

☒ 10

☐ 1.1

17

The control of free turbine engines on turboprops, is accomplished by: - a propeller control lever used to select: 1 - propeller RPM 2 - turbine temperature 3 - turbine RPM - a fuel control lever used to select: 4 - propeller RPM 5 - torque 6 - turbine temperature
The combination which regroups all of the correct statements is :

☐ 1 - 3 - 5

☐ 2 - 4 - 5

☒ 1 - 5 - 6

☐ 3 - 4 - 6

18

In a fire detection system with single-loop continuous components (with no fault protection), if the inner core contacts the outer casing:

- ☐ the main engine fire extinguisher is automatically activated.
- ☒ the fire alarm is triggered.
- ☐ there will be no effect on the system
- ☐ the power supply is cut off automatically.

19

When the oxygen system of a large aeroplane is used, :

- ☐ *the same source of supply is used by the cockpit crew and passengers.*
- ☐ *with the setting on "NORMAL", the passengers are supplied with a mixture of oxygen and cabin air.*
- ☐ *the same source of supply is used by the cockpit crew and cabin crew.*
- ☒ *with the setting on "NORMAL", the cockpit crew is supplied with a mixture of oxygen and cabin air .*

20

A passenger emergency mask is a :

- ☐ mask with flow on request and can be used if there is smoke.
- ☒ mask with flow on request and cannot be used if there is smoke in the cabin.
- ☐ continuous flow mask and can be used if there is smoke in the cabin.
- ☐ continuous flow mask and cannot be used if there is smoke in the cabin.

Candidate: Simon 2 Ekwue
Date of birth: 1987-08-29

Subject: 020-AGK
Date: 2014-07-17

21

(Refer to Figure 87.) At STRUT intersection headed eastbound, ATC instructs you to hold west on the 10 DME fix west of LCH on V306, standard turns, what entry procedure is recommended?

☒ **Direct.**

☐ **Parallel.**

☐ **Teardrop.**

See Attachments:

1. instrument_87.jpg

figure 87

(Attachment No.1)

22

How can a pilot identify a lighted heliport at night?

- ☒ White and red beacon light with dual flash of the white.
- ☐ Green and white beacon light with dual flash of the white.
- ☐ Green, yellow, and white beacon light.

23

If a pitot tube is clogged, which instrument would be affected?

- ☐ Altimeter.
- ☐ Vertical speed indicator.
- ☒ Airspeed indicator.

24

What should be the indication on the magnetic compass as you roll into a standard rate turn to the right from a south heading in the Northern Hemisphere?

- ☐ The compass will indicate a turn to the right, but at a faster rate than is actually occurring.
- ☐ The compass will remain on south for a short time, then gradually catch up to the magnetic heading of the airplane.
- ☒ The compass will initially indicate a turn to the left.

25

How can a pilot identify a military airport at night?

- ☐ White and red beacon light with dual flash of the white.
- ☐ Green, yellow, and white beacon light.
- ☒ Green and white beacon light with dual flash of the white.

26

When making an airborne VOR check, what is the maximum allowable tolerance between the two indicators of a dual VOR system (units independent of each other except the antenna)?

- ☒ 4 degrees between the two indicated bearings of a VOR.
- ☐ 6 degrees between the two indicated radials of a VOR.
- ☒ Plus or minus 4 degrees when set to identical radials of a VOR.

27

Which is the maximum tolerance for the VOR indication when the CDI is centered and the aircraft is directly over the airborne checkpoint?

- ☐ Plus 6degrees or minus 4 degrees of the designated radial.
- ☐ Plus or minus 4 degrees of the designated radial.
- ☒ Plus or minus 6 degrees of the designated radial.

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28

Which instrument would be affected by excessively low pressure in the airplane's vacuum system?

☐ Pressure altimeter.

☐ Airspeed indicator.

☒ Heading indicator.

29

A VSI may suffer from error due to:

- ☒ **pitot blockage, static blockage, aircraft manoeuvre**
- ☐ **time lag, instrument calibration, blockage**
- ☐ **density, aircraft manoeuvre, position of the static vent**
- ☐ **compressibility, time lag, blockage**

30

Given:

Pt = total pressure

Ps = static pressure

Pso = static pressure at sea level

The CAS is a function of:

☐ $(Pt - Pso) / Ps$

☐ $Pt - Ps$

☒ Pt / Ps

☐ $Pt - Pso$

31

Transition altitude is obtained from:

- ☐ ATC or VOR ATIS.
- ☒ 1500 feet above airfield elevation.
- ☐ Jeppesen or Aerad flight guides.

32

The QFE at an aerodrome (elevation) 1790 feet is 962 hPa and the QNH 1022 hPa. If the transition level is FL040 the physical level of the transition level above the aerodrome is approximately:

☐ 2464 feet

☒ 2280 feet.

☐ 2150 feet

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33

If the Pitot opening is blocked, which instruments would be affected (separate static vent)?

☐ ASI, Altimeter and VSI.

☒ ASI only.

☐ ASI and VSI.

34

If an aircraft climbs at a constant IAS in the standard atmosphere, the Mach number will:

- ☒ remain constant
- ☐ decrease
- ☐ increase
- ☐ remain constant unless the aircraft flies through an inversion

35

If an aircraft is descending at a constant Mach number in the standard atmosphere, the IAS will:

- ☐ decrease
- ☐ remain constant unless the aircraft flies through an inversion
- ☐ increase
- ☒ remain constant

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36

The following figures appear on an aeroplane deviation card: **FOR MAGNETIC (45----90), STEER COMPASS (38----92)**. The compass heading to steer to maintain a heading of **075° (M)** is:

☐ 74

☒ 72

☐ 76

37

The rotor of the DGI spins up and away from the pilot when 090 is indicated. The latitude compensation nut situated on the near right hand side of the inner gimbal from the gyro axis, has been set to give zero drift on the ground at the equator. To compensate for earth rotation at 30 S the latitude compensating nut:

☒ must be adjusted inwards;

☐ must be adjusted outwards;

☐ is not adjusted since the latitude nut can only be used to correct for apparent wander in the northern hemisphere.

38

At a low bank angle, the measurement of rate-of-turn actually consists in measuring the:

☒ *pitch rate of the aircraft.*

☐ *bank of the aircraft.*

☐ *yaw rate of the aircraft.*

☐ *roll rate of the aircraft.*

39

The horizontal command bar of a flight director:

- 1 - repeats the position information given by the ILS in the horizontal plane
- 2 - repeats the position information given by the ILS in the vertical plane
- 3 - gives information about the direction and the amplitude of the corrections to be applied on the pitch of the aircraft.

The combination that regroups all of the correct statements is:

- ☐ 2.
- ☐ 1, 3.
- ☒ 3.
- ☐ 2, 3.

40

The stall warning computer uses the following data:

- 1- mass of the aeroplane
- 2- angle of attack
- 3- wing flap deflection
- 4- position of the landing gear
- 5- TAT
- 6- pressure altitude

The combination that regroups all of the correct statements is:

☐ 2, 3, 5

☐ 1, 2, 3, 4, 5, 6

☐ 1, 2, 3, 4

☒ 2, 3

Attachment No. 1

Questions 21

figure 87

instrument_87.jpg

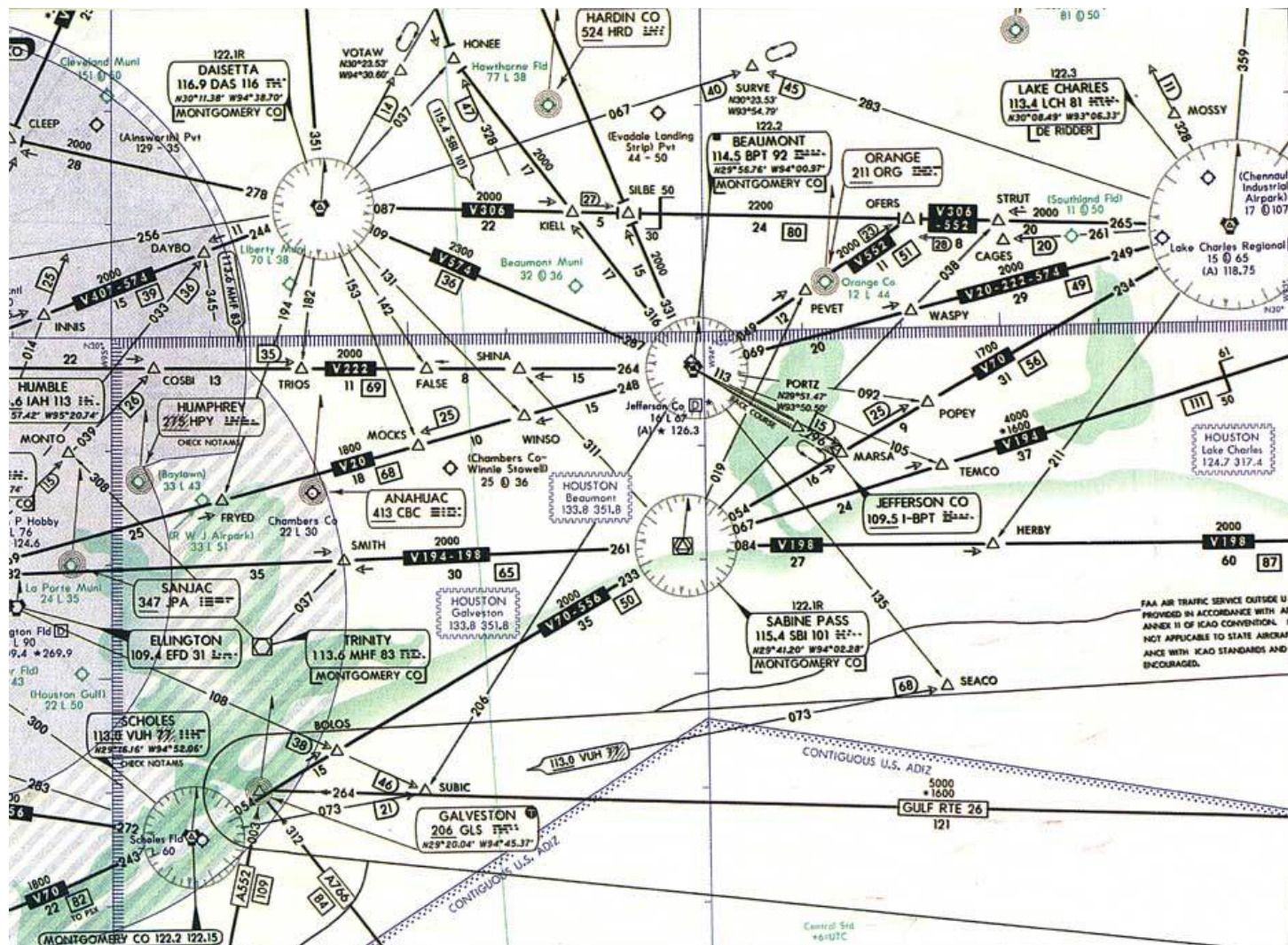


Figure 87. En Route Chart Segment © ASA